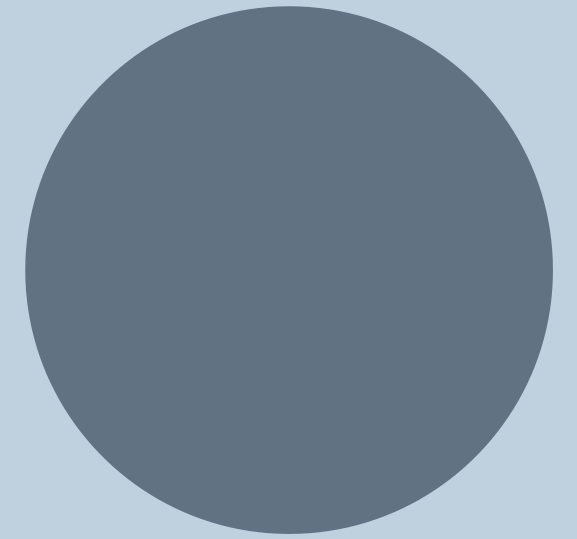
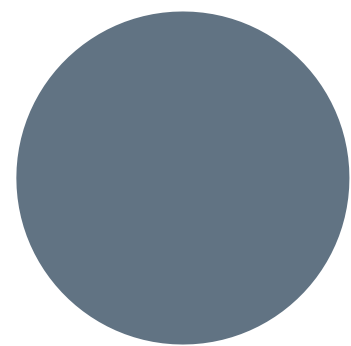

MicroProcessors & Microcontrollers

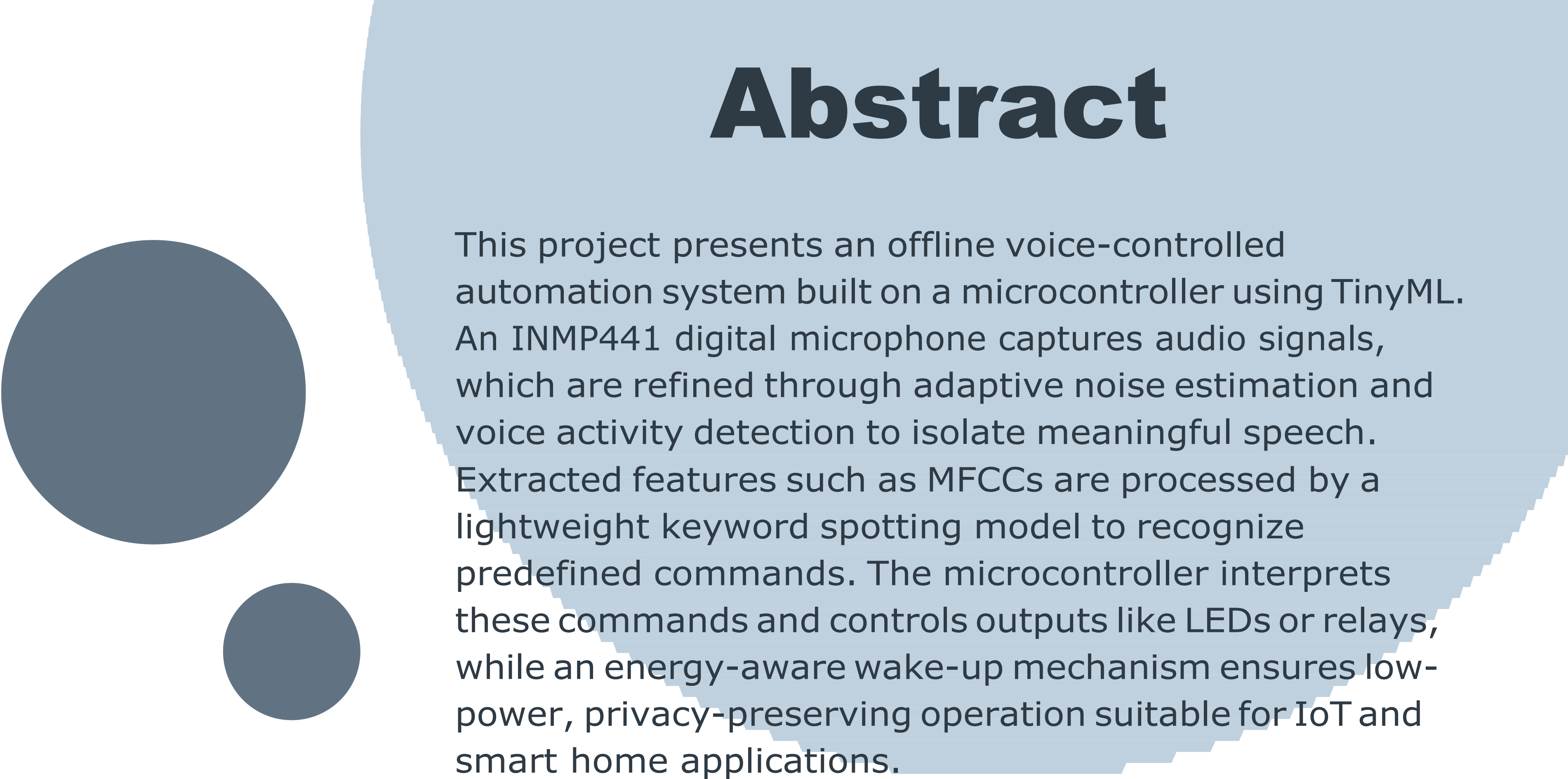


Introduction

Adaptive Noise-Robust Offline Voice-Controlled Microcontroller using TinyML

Voice-controlled systems are becoming common in the modern technological landscape. Most of the systems rely on cloud and high performance processors. This makes them unsuitable for low-power environments such as a microcontroller based system. However, there has been advancements in the area of TinyML which enables us to run Machine Learning models directly on microcontrollers. This makes it possible for intelligent signal processing at the level of edge. In this project we implement an adaptive noise-robust offline voice-controlled microcontroller system using TinyML.





Abstract

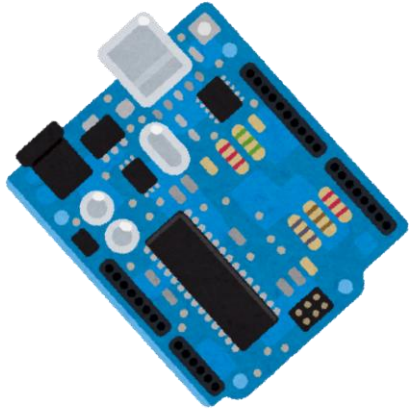
This project presents an offline voice-controlled automation system built on a microcontroller using TinyML. An INMP441 digital microphone captures audio signals, which are refined through adaptive noise estimation and voice activity detection to isolate meaningful speech. Extracted features such as MFCCs are processed by a lightweight keyword spotting model to recognize predefined commands. The microcontroller interprets these commands and controls outputs like LEDs or relays, while an energy-aware wake-up mechanism ensures low-power, privacy-preserving operation suitable for IoT and smart home applications.

Problem Statement

Voice-controlled systems are increasingly common, but most rely on cloud services and high-performance processors, making them unsuitable for low-power microcontroller environments. This dependence introduces latency, privacy risks, and energy inefficiency. With the rise of TinyML, machine learning models can now run directly on microcontrollers, enabling intelligent signal processing at the edge. However, ensuring robust performance in noisy conditions remains a challenge. Thus, the problem is to design an adaptive, noise-robust, offline voice-controlled system using TinyML on microcontrollers, providing low-power, real-time, and reliable operation without reliance on external servers.



Components



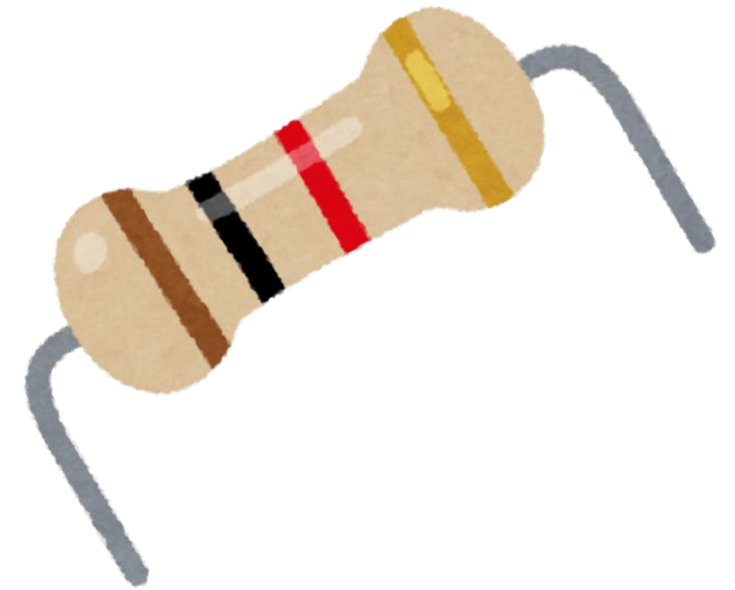
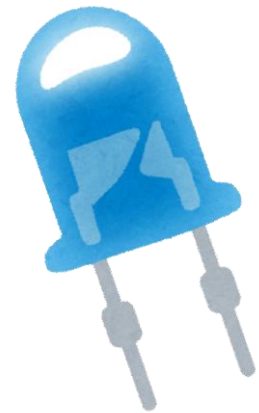
ESP32 / Nano 33 BLE

INMP441 Microphone

LED / Relay / Motor

5V USB / Battery Supply

Breadboard with Jumper Wires



Project goal

- To develop a voice command recognition system that operates without depending on the cloud.
To deploy a TinyML keyword spotting model.
Enable a compact and cost-effective edge intelligent system.

- To incorporate an adaptive noise-handling mechanism in a scalable architecture.
Prioritizing privacy, low-latency and energy efficiency and processing at an edge level.

BLOCK DIAGRAM

Microphone INMP441

Adaptive Noise & VAD

TinyML Voice Recognition

MCU Control

Output Module



Applications

- Smart Home Devices: Control lights, fans, or appliances in real time without internet dependency, ensuring privacy and low power consumption.
- Healthcare Assistive Tools: Voice-activated aids for elderly or differently abled individuals, functioning reliably even in noisy hospital or home environments.
- Industrial Automation: Hands-free machine operation in factories where background noise is high, improving worker safety and efficiency.
- Wearable Electronics: Smartwatches or fitness trackers with offline voice commands, reducing reliance on smartphones or cloud connectivity.
- Agricultural IoT Systems: Farmers can control irrigation pumps or monitoring devices through voice commands in rural areas with limited connectivity.
- Automotive Interfaces: In-vehicle voice control for basic functions (windows, AC, infotainment) that works offline and adapts to road noise.
- Educational Tools: Low-cost, voice-interactive learning kits for students, especially in resource-constrained environments.

This project essentially bridges the gap between intelligent voice interfaces and low-power embedded systems, making them viable in everyday, connectivity-limited scenarios.

Existing Works

Patent: Noise robust representations for keyword spotting systems(US12027156B2 / US20230104431A1)

This patent discloses methods for improving keyword spotting (KWS) systems so they are more robust to noise and speaker variations. That means the system is designed to better detect specified keywords (wake-words or command words) in audio even when the environment contains background noise.

Patent: Low-power keyword spotting system(US20230409102A1)

Describes a keyword spotting system that operates with low power and detects wake-words before triggering higher-level speech recognition.

Patent: Embedded system for construction of small footprint speech recognition with user-definable constraints (US9117449B2)

Describes an embedded system for voice trigger/wakeup that runs on a secondary low-power processor or DSP, not on a full main CPU..



Novelty

- Unlike existing systems that rely on static detection thresholds, the project implements an adaptive noise-robust mechanism in its firmware logic that continuously estimates ambient acoustic conditions and dynamically adjusts voice detection sensitivity, improving reliability in real-world environments.
 - The system deploys a TinyML-optimized neural network model directly on resource-constrained hardware, enabling fully offline operation with low memory and power requirements while preserving responsiveness and user privacy.
-

Conclusion

The project demonstrates an offline voice-controlled system where adaptive noise thresholding and TinyML-based keyword recognition are integrated on an ESP32. Ambient noise is handled through firmware-level frame energy analysis, while the neural model performs lightweight classification, resulting in efficient, privacy-preserving, and robust operation on low-power hardware. This validates the practicality of deploying intelligent voice interfaces on edge microcontroller platforms.

References

<https://patents.google.com/patent/US20240347057A1/en>

Competing devices responding to voice triggers

At a first electronic device with a display and a microphone: sampling audio input using the first microphone; in accordance with the sampling of audio input using the first microphone, sending stop instructions to a second electronic...

 google.com / Aug 31, 2015

Noise robust representations for keyword spotting systems

Described are techniques for noise-robust and speaker-independent keyword spotting (KWS) in an input audio signal that contains keywords used to activate voice-based human-computer interactions. A KWS system may combine the latent...

 google.com / Feb 22, 2022



**Thank
You**